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First High-Resolution Tomographic View of the Circumgalactic Medium

The $z = 0.73$ Mg II - Fe II absorber toward the Sunburst Arc

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ABSTRACT

We present the first high spectral resolution tomographic view of a circumgalactic medium (CGM) absorber, using the brightest known gravitationally lensed arc, the 'Sunburst Arc'. We study an intervening Mg II - Fe II absorption system at $z \sim 0.73$ associated with an identified galaxy, using UVES/VLT spectroscopy of eight spatially resolved sight-lines probing projected distances of 3 - 28 kpc along the galaxy's major axis.

We detect strong Mg II absorption across the lensed arc, with a covering fraction close to unity. We model the Mg II and Fe II absorption lines in each pseudo-spaxel using single component Voigt profiles at a spectral resolution of $R \simeq 30,000$. Despite the high spectral resolution, we do not resolve multiple velocity components, and instead find large Doppler parameters. These broad profiles are likely produced by the combined velocity dispersion of multiple unresolved absorbing components within each pseudo-spaxel.

Comparison with quasar sight-lines and simulated pseudo-spaxel observations shows that recovered column densities are affected by both Mg II saturation and spatial averaging. From the data, the increase in the Mg II/Fe II ratio toward larger galactocentric distances along the major axis is qualitatively consistent with an α-enhanced outer CGM, although ionization and dust depletion may also contribute.

These results demonstrate the potential of high resolution tomography with gravitational arcs for studying the spatial, kinematic, and chemical structure of the CGM on small physical scales.

Key words. galaxies: clusters: individual: PSZ1 G311.65–18.48 – galaxies: evolution

1. Introduction

1. Developing a complete picture of galaxy evolution requires understanding the role of the circumgalactic medium (CGM), which serves as the interface between galaxies and their surrounding environment (Tumlinson et al. 2017; Péroux & Howk 2020; Faucher-Giguère & Oh 2023). Over the past decades, absorption-line spectroscopy of background quasars has become one of the primary tools for probing the CGM, providing detailed constraints on its physical, chemical, and kinematic properties (Nielsen et al. 2013; Huang et al. 2021) . However, each quasar probes the halo along only a single pencil-beam sight-line, limiting our ability to investigate the spatial distribution, kinematic, and small-scale structure of the absorbing gas within individual halos.
2. A more direct view of the CGM can be obtained through multiple sight-lines intersecting the same halo, allowing spatial variations in absorber properties to be measured within individual galaxies (Steidel et al. 2010; Rubin et al. 2018). Strong gravitational lensing provides an ideal opportunity by producing multiple closely separated sight-lines through the same foreground halo, making it an ideal tool for tomographic studies of the CGM. (e.g., Lopez et al. 2018, 2020, 2024; Tejos et al. 2021; Mortensen et al. 2021; Fernández-Figueroa et al. 2022; Afruni et al. 2023; Shaban et al. 2025; Hernández-Guajardo et al. 2026).
3. Most existing tomographic studies have relied on low- or intermediate-resolution spectroscopy. Although these observations have revealed the large-scale distribution of CGM absorbers, they can not fully resolve their internal velocity structure. High-resolution spectroscopy of spatially extended background sources can provide access to the detailed kinematics, ionization structure, and small-scale variations of the absorbing gas while simultaneously preserving spatial information across multiple sight-lines. The feasibility and scientific potential of this approach were demonstrated by Diamond-Stanic et al. (2016), who used high-resolution spectroscopy of an extended background source to investigate intervening absorption systems at $z = 0.4$.
4. In this work, we use the unique geometry of the brightest known strongly lensed arc, the 'Sunburst Arc' (see Fig. 1), located behind the galaxy cluster PSZ1 G311.65–18.48 (Dahle et al. 2016). The arc is produced by strong gravitational lensing of a star-forming galaxy at $z = 2.369$ by a massive cluster at $z = 0.443$, providing multiple closely separated sight-lines through intervening absorbers. An intervening Mg II–Fe II absorption system at $z = 0.73379$ was identified by Lopez et al. (2020) toward the northernmost portion of the arc. Nebular [O II] emission detected at the same redshift reveals a nearby galaxy, which we identify as the absorber (hereafter 'G1'). Using this system, we aim to investigate the spatial distribution, kinematics, and chemical properties of low-ionization gas on kiloparsec scales.